SOFTWARE TESTING
CO3015 / CO5252

CH1. INTRODUCTION

CONTENT

- The role and the importance of software testing
- Testing levels
- Testcases
- Basic principles of software testing
- Testing process and plan
- Test automation

WHY DO WE TEST SOFTWARE?

- Software is with us!
  - Is it safe?
- Fault vs. Failure vs. Error
  - Fault : It is a condition that **causes** the software to fail to perform its required function.
  - Error : Refers to **difference** between Actual Output and Expected output.
  - Failure : It is the **inability** of a system or component to perform required function according to its specification.

https://softwaretestingtimes.com/2010/04/fault-error-failure.html

WHAT IS SOFTWARE TESTING?

- Software Testing is **a method to check whether the actual software product matches expected requirements and to ensure that software product is defect free**.
  - It involves execution of software/system components using manual or automated tools to evaluate one or more properties of interest.
  - The purpose of software testing is to identify errors, gaps or missing requirements in contrast to actual

- [2] Testing is the process of executing a program with the **intent of finding errors**.

https://www.guru99.com/software-testing-introduction-importance.html

THE ROLE AND THE IMPORTANCE OF SOFTWARE TESTING

- The **role** of testing in software development begins with **improved reliability, quality and performance of the software**.
- It assists a developer to check out whether
  - the software is **performing the right way**
  - and to assure that software is **not performing what it is not supposed to do**.

https://www.testbytes.net/blog/role-of-software-testing-in-software-development/

TESTCASES

- A test case is **a specification of the inputs, execution conditions, testing procedure, and expected results that define a single test** to be executed to achieve a particular software testing objective, such as to exercise a particular program path or to verify compliance with a specific requirement

https://en.wikipedia.org/wiki/Test_case

TESTCASES VS. TEST SCENARIO

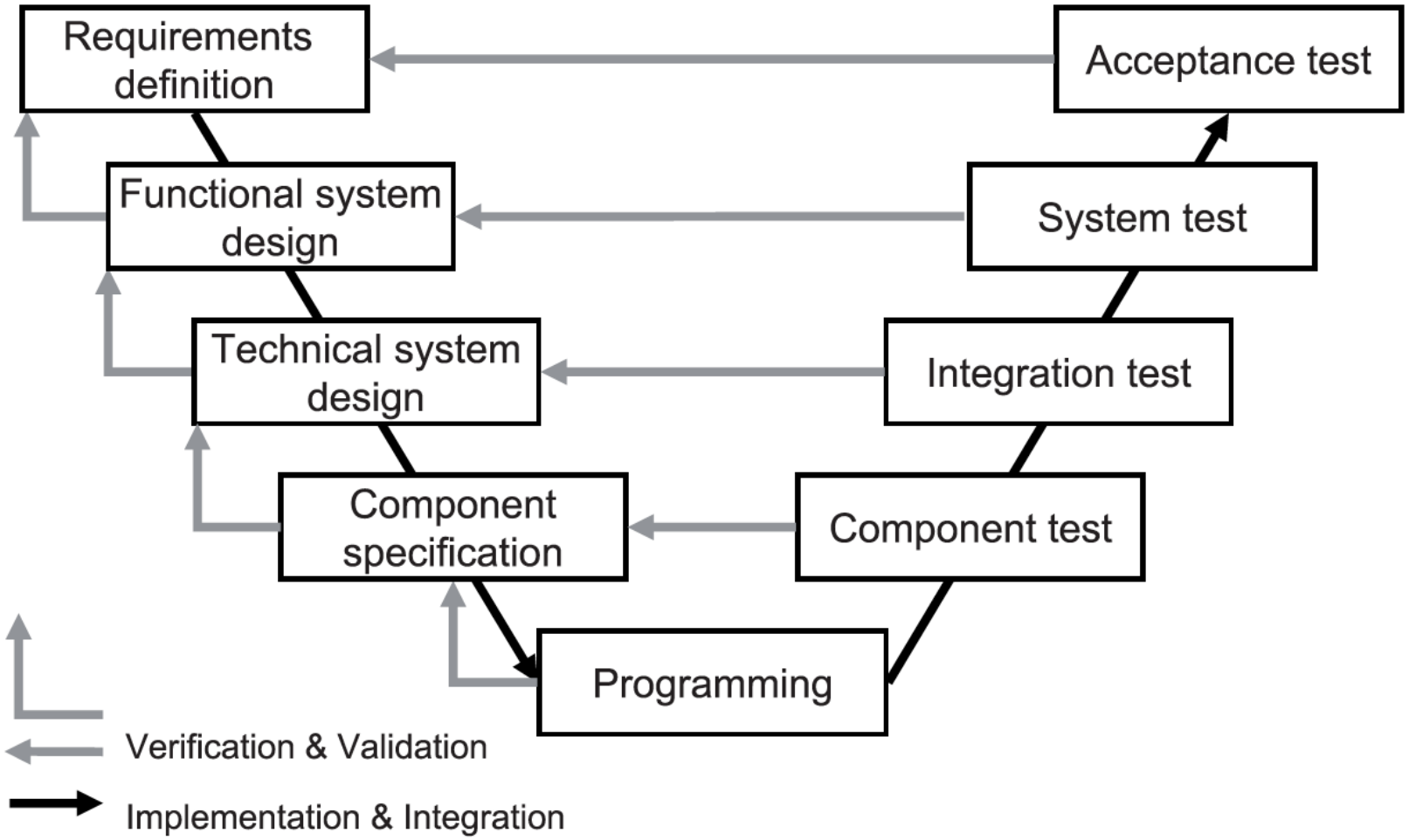
- A Test case is a set of actions executed to verify a particular feature or functionality of your software application.
  - A Test Case contains **test steps, test data, precondition, postcondition** developed for specific test scenario to verify any requirement. The test case includes specific variables or conditions, using which a testing engineer can **compare expected and actual results** to determine whether a software product is functioning as per the requirements of the customer.
- A Test Scenario is defined as **any functionality that can be tested**. It is a collective set of test cases which helps the testing team to determine the positive and negative characteristics of the project.

https://www.guru99.com/test-case-vs-test-scenario.html

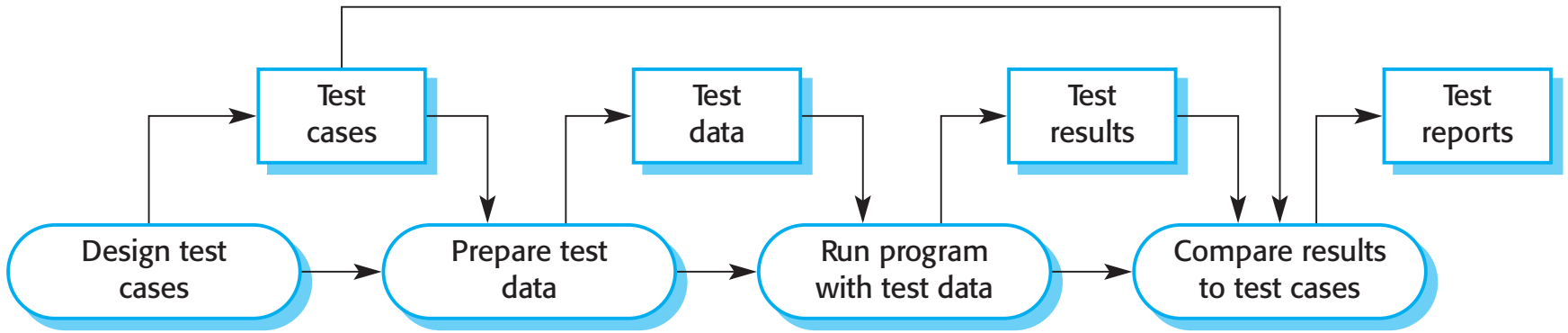
BASIC PRINCIPLES OF SOFTWARE TESTING [3]

- Principle 1: Testing **shows** the presence of defects, **not their absence**.
- Principle 2: **Exhaustive** testing is **impossible**.
- Principle 3: Testing activities should **start** as **early** as possible.
- Principle 4: Defect **clustering**.
- Principle 5: The **pesticide paradox**.
- Principle 6: Testing is **context dependent**.
- Principle 7: **No failures** means the **system is useful is a fallacy**.

TESTING PROCESS – THE GENERAL V-MODEL [3]

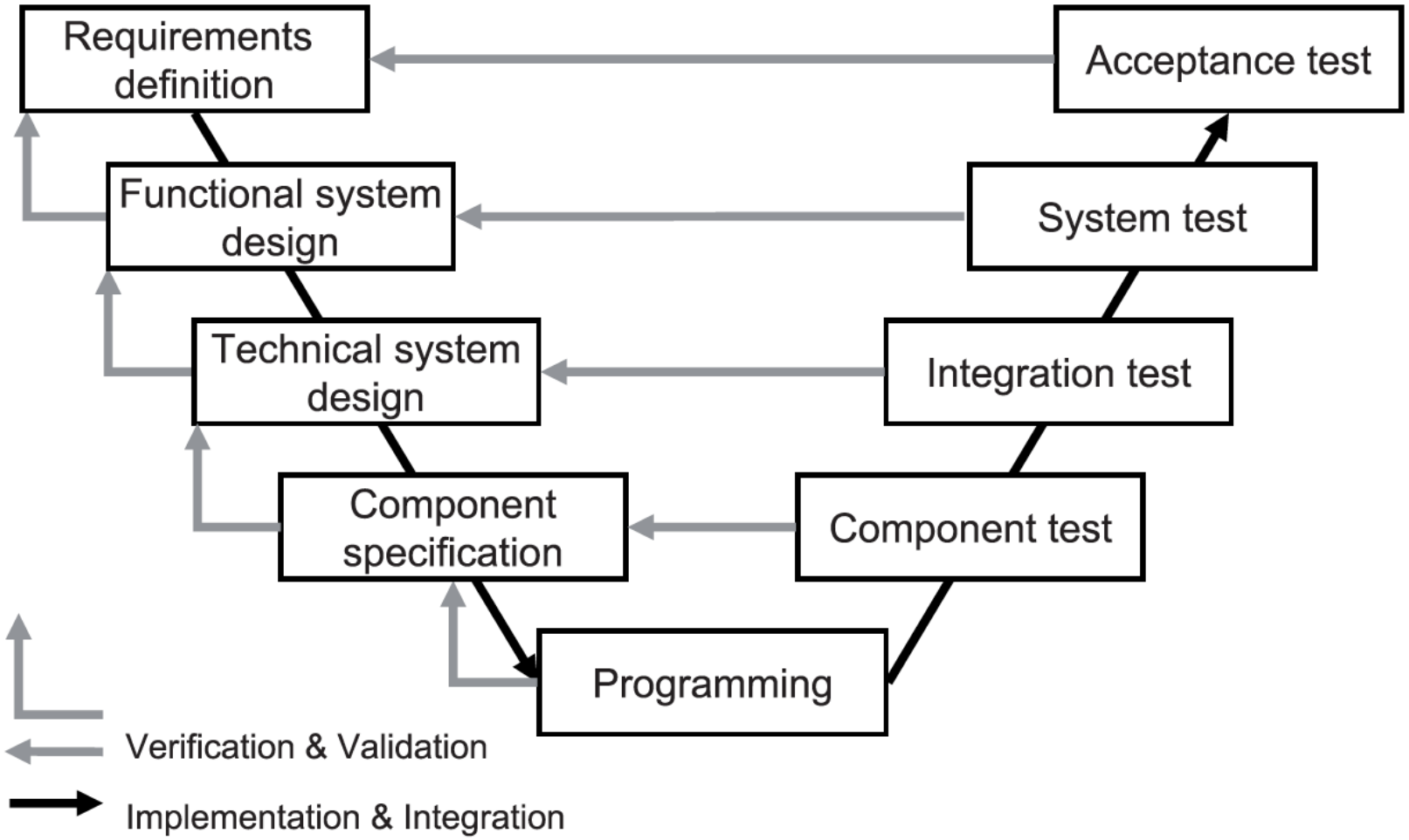


A MODEL OF THE SOFTWARE TESTING PROCESS



TESTING LEVELS

https://www.seguetech.com/the-four-levels-of-software-testing/

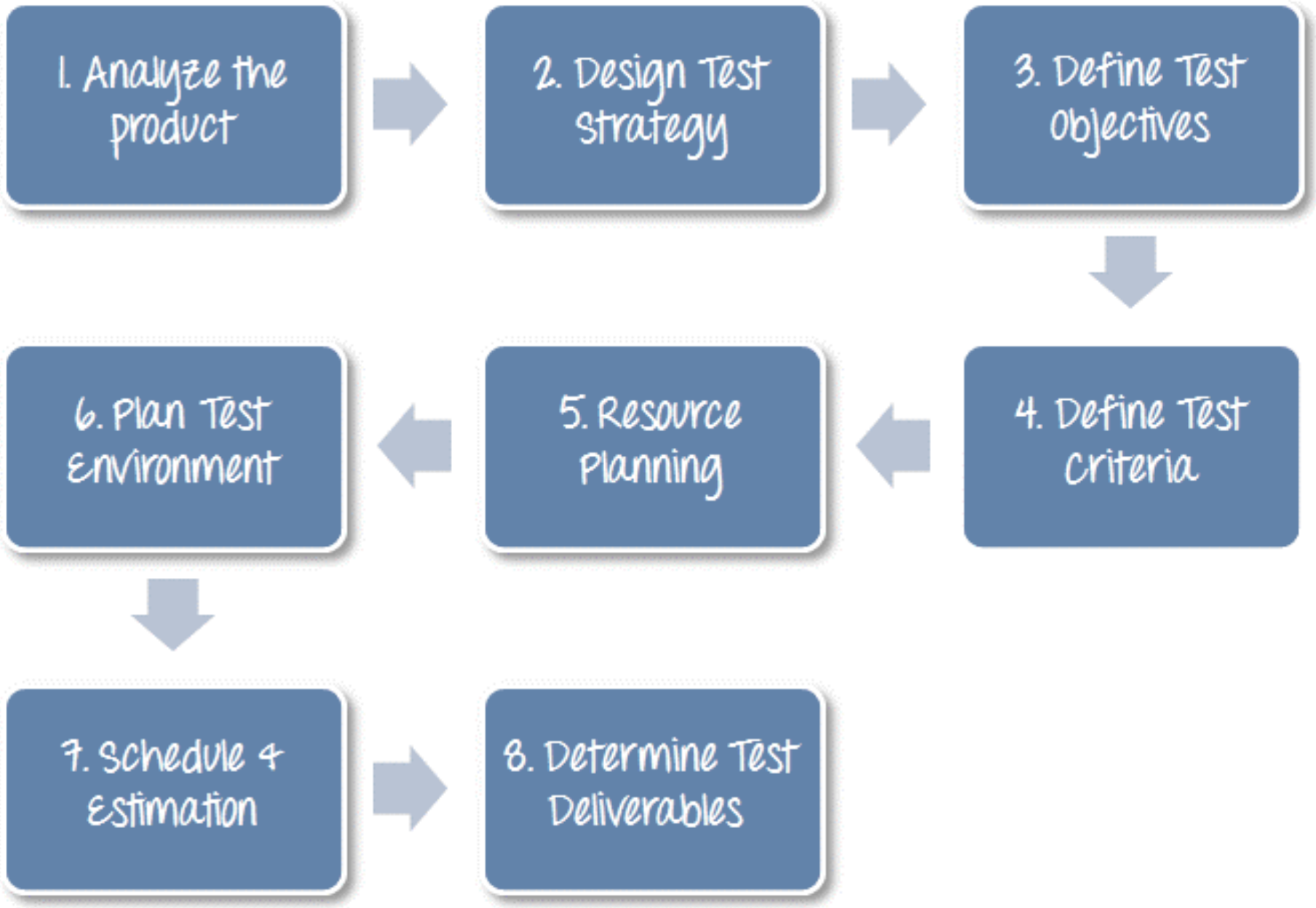


TEST PLAN

- A **Test Plan** is a detailed document that **describes** the test **strategy**, **objectives**, **schedule**, **estimation**, **deliverables**, and **resources** required to perform testing for a software product.
  - Helps us determine the effort needed to validate the quality of the application under test.
  - Serves as a blueprint to conduct software testing activities as a defined process, which is minutely monitored and controlled by the test manager.
- As per ISTQB definition:
  - "Test Plan is a document describing the **scope**, **approach**, **resources**, and **schedule** of intended test activities."

https://www.guru99.com/what-everybody-ought-to-know-about-test-planing.html

- A **Test P** … **ategy, objectiv** … **es** requirec
  - Helps u … lication under te
  - Serves a … process … er.
- As per IS
  - "Test Pl … and **schedu**



…-planing.html

TEST AUTOMATION

- The **use of software to control** the execution of tests, the comparison of actual outcomes to predicted outcomes, the setting up of test preconditions, and other test control and test reporting functions

SUMMARY

- Software testing is important
- What is software testing vs. testing purposes/goals
  - Satisfy the requirements vs. finding defects
- Testing levels
  - Unit/Component -> Integration -> System -> Acceptance
- Testcases vs. Test scenarios
- 7 basic principles of software testing
- Testing process and plan
- The V-model
- The plan
- Test automation